

QTD-HGNN - Business Case

Where it fits, why banks buy, how they integrate, scale & maintain

QTD-HGNN - Quantum-Topological Threat Correlation

Problem Statement 2: AI-Driven Correlation of Cybersecurity

Telemetry & Transactional Behaviour

1. The problem worth money

Banks run 10-15 siloed security tools (EDR, SIEM, IAM, fraud engines) that do not talk to each other. The security team watches logins; the fraud team watches payments. An account takeover - a hostile login followed 90 seconds later by a large transfer to a new payee - is invisible to each team alone. That seam is where losses happen.

Signal (all [Unverified] - from public reporting)	Figure
Bank frauds, FY25 (India) vs prior year	~3x increase
UPI fraud cases, FY25	~85% rise (13.4 lakh cases)
Mule accounts identified (India)	~5.24 lakh
False-positive rate in current SOCs	90%+ (alert fatigue)
Mean time to detect a breach (some sectors)	~200 days

Why now

RBI is actively pushing MuleHunter.AI and a 'zero fraud' target; the RBI Q-SAFE committee (2026) mandates a cryptographic inventory + PQC migration roadmap; fraud is at record highs. Adopting now is compliance, not a nice-to-have.

2. What we sell - and why they buy

An AI correlation + threat-intelligence OVERLAY that fuses cyber telemetry with transaction behaviour, catches what siloed tools miss, cuts false positives through cross-domain corroboration, and explains every alert for auditors.

The buying levers, in the order a CISO cares

- **1. Regulation:** It helps meet obligations the bank must meet anyway - RBI Fraud Risk Management Master Direction (2024) mandates data-analytics Early Warning Signals + real-time transaction monitoring; RBI Q-SAFE mandates a crypto inventory. We are a ready-made way to satisfy both. Selling compliance is easier than selling technology.
- **2. Loss reduction:** Attacks a bleeding-money problem - even a small reduction in fraud loss or in analyst hours wasted on false positives is a hard ROI number.
- **3. Efficiency:** Fewer, higher-confidence alerts cut SOC operating cost and analyst burnout.

3. Where it runs & where data lives

This is the question that decides viability for an Indian bank, so we answer it head-on.

- **Data residency:** It deploys INSIDE the bank's own cloud tenant (Azure, India region) - not our SaaS. Raw transaction data and PII never leave the bank's security boundary. This is mandatory under RBI data-localization + DPDP Act 2023, and it is the single most important architectural decision - driven by law, not preference.
- **Delivery:** Containerised (Docker -> Azure Kubernetes / Container Apps), deployed via an infrastructure-as-code template into the bank's subscription. Storage uses the bank's own database; compute is the bank's own cluster.
- **Isolation:** The vendor receives zero raw data. Cloud-prem / BYOC: vendor-managed control plane, data plane in the bank's VPC. Air-gapped installs for the most locked-down banks.

Reference: this exact product

Backend runs on Azure Container Apps in one resource group (ACR + app + Postgres/SQLite). Frontend on Vercel. CI/CD via GitHub Actions. Live, deployed, and streaming real model output today - the deployment model is proven, not hypothetical.

4. How it integrates (be a good citizen, not a rip-and-replace)

The product sits ON TOP of what the bank already owns and ingests from it.

Integration point	Mechanism
Network telemetry	Passive TAP / SPAN mirror port -> Zeek/Suricata sensors (out-of-band, nothing to break)
TLS / crypto posture	JA3/JA4 fingerprinting on ClientHello - flags weak/quantum-vulnerable ciphers WITHOUT decryption
SIEM	Ingest + push-back via syslog/CEF, Kafka, REST to Splunk / QRadar / Microsoft Sentinel
Core banking / payments	Feed from Finacle, payment switch, IAM via syslog / Kafka / DB export
Response	Fire SOAR playbooks (webhook/REST); alerts appear in the existing SOC queue
Threat intel	STIX 2.1 over TAXII 2.1 in/out

What makes it 'easy to integrate' for a buyer: prebuilt SIEM connectors, open schemas, TAXII in/out, and no requirement to reroute production traffic.

5. How it scales

- **Per-node:** The trained model is small and CPU-only (no GPU cluster). Inference is milliseconds; the heavy work is pre-computed offline at build time.
- **Per-bank:** Each bank runs its own isolated instance (per-tenant). You scale operationally by adding tenants, never by centralising anyone's data.
- **Cross-bank:** Banks opt into a federated threat-intelligence exchange: only ANONYMISED attack fingerprints are shared (mule-ring hashes, ATO patterns) - never customer data. Each new bank makes every bank's detection better. This is the network-effect moat, and it mirrors the real I4C Suspect Registry model (share indicators, not data).

Honest scope note

Federated learning across banks is on the roadmap (research-grade today). Shipping now: per-tenant deployment + IOC/TAXII indicator sharing, which is proven production practice.

6. How it is operated & maintained

- **Day-to-day:** Slots into the bank's mandated 24x7 SOC: L1 analysts triage the alert queue, escalate to L2/L3 as cases, drive response via SOAR.
- **MLOps:** Model retraining, drift monitoring, and false-positive tuning - run on the bank's own data on a schedule; jointly owned by the SOC and vendor support.
- **Upkeep:** Push-to-deploy CI/CD; model weights hot-swappable; migrations automated. Explainable outputs give a permanent audit trail (every alert decomposes into SHAP reason codes).

7. Where it fits & the market

- **Fit:** An intelligence overlay between the SOC (security) and the fraud/EFRRMS engine - the seam neither currently owns. It does not replace Splunk, QRadar, or the fraud engine.
- **Market:** RBI-regulated banks (public & private), co-operative banks and RRBs (underserved by enterprise vendors), SWIFT participants, insurance, capital markets - anyone facing the fraud + quantum-risk mandate.

Adoption path

Phase 1 (M1-3): pilot in a single SOC, passive TAP/SPAN PoC. Phase 2 (M4-6): SIEM connector + fraud-feed integration. Phase 3 (M7-9): multi-branch rollout. Phase 4 (M10-12): full production + PQC-posture monitoring.

8. Commercial model

- **Core:** Annual license per deployment, tiered by transaction volume / branch count.
- **Compliance add-on:** Q-SAFE crypto-inventory + audit-reporting module as a separate SKU (maps to a deadline -> high margin).
- **Network:** Federated-intelligence subscription - recurring, and the moat.
- **Growth:** Expansion: SWIFT participants, insurance, capital markets, and export to other emerging-market regulators with similar data-residency laws.

The one-line pitch

The only layer that sees the crypto-vulnerability of a data movement AND its transactional context in one place - answering 'was this bulk KYC export protected by quantum-breakable crypto, and did it have any legitimate business reason?' Neither the bank's EFRMS, nor its SIEM, nor a generic crypto tool can ask that question.

9. Honesty & compliance posture

Every dashboard number is computed by the trained model, not simulated. The quantum module reports real cryptographic posture and observable HNDL indicators; it does not claim to detect passive interception, which is physically undetectable - the mitigation is PQC migration. Data is synthetic today, with an ingestion format designed to swap in real bank feeds. This honesty is a selling point: it survives technical due diligence and satisfies RBI/DPDP/EU-AI-Act transparency expectations.